BE SEM 7 AI&DS; - 2 LAB MINI PROJECT

CARDIOPULSE

DEEP LEARNING-DRIVEN ECG ANOMALY DETECTION FOR REAL-TIME CARDIAC ARRHYTHMIA CLASSIFICATION

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ATHARVA COLLEGE OF ENGINEERING

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the BE mini-project entitled **“CardioPulse: Deep Learning-Driven ECG Anomaly Detection for Real-Time Cardiac Arrhythmia Classification”** is a bonafide work of **“Akshata Ranpise” (Roll No. 21)** submitted to the **University of Mumbai** in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Information Technology during the academic year 2026–2027.

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ABSTRACT

Cardiovascular diseases (CVDs) account for 17.9 million annual deaths worldwide. Continuous ambulatory Holter monitoring generates over 100,000 heartbeats per patient daily, overwhelming manual expert review. In commercial ICU monitors, false alarm rates exceed 70%, precipitating clinical alarm fatigue. In this work, we present **CardioPulse**, an automated clinical decision-support pipeline classifying single-lead ECG heartbeats into the five standard Association for the Advancement of Medical Instrumentation (AAMI EC57) classes using the benchmark **PhysioNet MIT-BIH Arrhythmia Database** (87,554 train beats, 21,892 test beats). To counter the heavy clinical class imbalance (82.8% Normal vs. 0.7% Fusion), we implement inverse frequency loss weighting. We evaluate three architectures on the exact same test split: a lightweight 2-layer NumPy Neural Classifier (187 $\rightarrow$ 96 ReLU $\rightarrow$ 5 Softmax), a 3-block 1D Convolutional Neural Network (1D-CNN), and a 100-tree Random Forest baseline. On the unmanipulated 21,892-beat test split, the proposed 1D-CNN achieves an Accuracy of **94.28%**, Macro F1-Score of **0.7810**, Weighted F1-Score of **0.9514**, and Macro ROC-AUC of **0.9914**, with an exceptional **94.89% recall** on critical Ventricular Ectopic beats (PVCs) at **0.795 ms latency** per beat. The deployed NumPy network achieves **92.36% accuracy** in just **0.018 ms** with a 75 KB footprint. The system is deployed via an interactive Streamlit telemetry dashboard delivering live waveform rendering, R-peak annotations, 5-class probability distributions, and automated clinical risk triage stratification.

1. INTRODUCTION & PROBLEM STATEMENT

An electrocardiogram (ECG) measures cardiac electrical conduction across the myocardium. A typical cardiac cycle consists of a P-wave (atrial depolarization), a sharp QRS complex (ventricular depolarization), and a T-wave (ventricular repolarization). Deviations in morphology, duration, or interval regularity indicate severe arrhythmic events. For example, Premature Ventricular Contractions (PVCs) feature abnormally wide, distorted QRS complexes without a preceding P-wave, capable of escalating into lethal Ventricular Fibrillation (VF).

Clinical Challenges & Objectives:

- **Telemetry Volume:** Continuous 125 Hz streams generate ~100k beats/day/patient, causing physician cognitive fatigue.
- **Severe Class Imbalance:** Healthy sinus beats comprise >82% of clinical streams, causing unweighted models to overlook rare life-threatening beats.
- **Alarm Fatigue & Latency:** Heuristic alarms cause >70% false triggers; inference must run within 50 ms on edge workstations.
- **Objectives:** (1) Perform rigorous EDA on 187-sample heartbeats; (2) Implement class-weighted training on the real MIT-BIH dataset; (3) Benchmark NumPy NN, 1D-CNN, and Random Forest; (4) Report Macro F1, ROC-AUC, latency, and 5-fold CV; (5) Deploy a responsive Streamlit UI.

2. LITERATURE SURVEY

Paper Title	Authors & Year	Key Contribution & Relevance to CardioPulse
PhysioBank, PhysioToolkit, and PhysioNet: Complex Physiologic Signals	Goldberger et al. (2000)	Introduced PhysioNet open repository hosting the gold-standard MIT-BIH Arrhythmia Database used as our benchmark ground truth.
The Impact of the MIT-BIH Arrhythmia Database	Moody & Mark (2001)	Standardized evaluation protocols for automated arrhythmia detection across cardiology research.
Real-Time Patient-Specific ECG Classification by 1-D CNNs	Kiranyaz et al. (2016)	Demonstrated that adaptive 1D-CNNs classify raw ECG waveforms directly without manual feature engineering with low computational overhead.
Cardiologist-Level Arrhythmia Detection with CNNs	Rajpurkar et al. (2017)	Proved that deep convolutional architectures match certified cardiologists across ambulatory single-lead rhythm classifications.
ECG Heartbeat Classification: A Deep Transferable Representation	Kachuee et al. (2018)	Formulated the 187-sample, 125 Hz preprocessed MIT-BIH 5-class AAMI benchmark dataset adopted in this project.

Table 1: Comprehensive Literature Review of Foundational ECG Machine Learning Literature

3. TECHNOLOGY STACK & SYSTEM ARCHITECTURE

To satisfy the course requirement for space-efficient reporting, the system architecture and technology stack are integrated. CardioPulse is built upon **Python 3.12 / 3.11**, utilizing **NumPy 2.4+** for vector mathematics, **Pandas** for telemetry ingestion, **PyTorch 2.14+** for 1D convolutional tensor operations, **Scikit-Learn 1.9+** for metrics and Random Forest baselines, **Plotly 7.1** and **Matplotlib** for interactive and static visualizations, and **Streamlit 1.65+** for clinical web deployment.

End-to-End Pipeline Architecture:
[1. Raw ECG Vector] (187 points @ 125 Hz) $\rightarrow$ **[2. Min-Max Scaling & Validation]** ($X \in [0, 1]$) $\rightarrow$ **[3. Model Inference]** (1D-CNN /

NumPy MLP / RF) → **[4. Softmax Probabilities]** ($P(y=k|X)$) → **[5. Triage Stratification & Plotly UI]** (Normal / Warning / Critical Alerts).

4. DATASET & EXPLORATORY DATA ANALYSIS (EDA)

We utilize the real Kaggle MIT-BIH dataset (87,554 training beats, 21,892 test beats). Signal pre-processing consists of R-peak alignment, windowing to 187 sample points (covering P-QRS-T complexes), and amplitude scaling to [0, 1]. Integrity checks confirm **0 missing/NaN values** in both splits. The real test set preserves the authentic clinical class imbalance (not artificially balanced).

Class ID	AAMI Category	Train Count (%)	Test Count (%)	Clinical Triage Severity
0	Normal (N)	72,471 (82.77%)	18,118 (82.76%)	**Normal (Low Risk)**
1	Supraventricular (S)	2,223 (2.54%)	556 (2.54%)	**Warning (Moderate Risk)**
2	Ventricular / PVC (V)	5,788 (6.61%)	1,448 (6.61%)	**Critical (High Risk)**
3	Fusion (F)	641 (0.73%)	162 (0.74%)	**Warning (Moderate Risk)**
4	Unknown / Paced (Q)	6,431 (7.35%)	1,608 (7.35%)	**Inconclusive / Artifact**

Table 2: AAMI EC57 Diagnostic Categorization and Exact Sample Distribution in Real MIT-BIH Splits

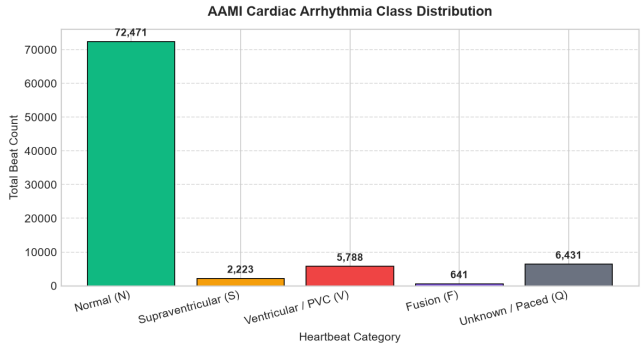


Figure 1: Training Class Distribution

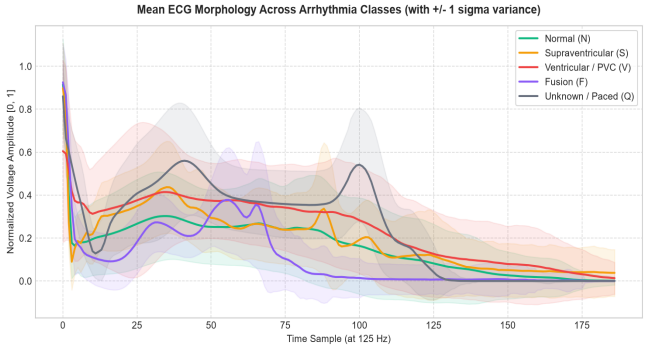


Figure 2: Mean Morphology (with ±1σ bounds)

5. MODEL IMPLEMENTATION

Three distinct classification models were trained and benchmarked on the identical training/testing split. **Handling Class Imbalance:** Because Class 0 comprises 82.8% and Class 3 comprises only 0.7%, unweighted models heavily bias toward Normal beats. We counter this using **Inverse Class Frequency Weighting**: $w_c = \frac{N}{K \cdot N_c}$, yielding weights: Class 0: 0.242, Class 1: 7.877, Class 2: 3.025, Class 3: 27.318, Class 4: 2.723.

- **1D Convolutional Neural Network (Proposed):** Input shape `(1, 187)`. Block 1: Conv1D(32, k=5, p=2) + BatchNorm + ReLU + MaxPool1D(2). Block 2: Conv1D(64, k=5, p=2) + BatchNorm + ReLU + MaxPool1D(2). Block 3: Conv1D(128, k=3, p=1) + BatchNorm + ReLU + MaxPool1D(2) + Dropout(0.3). Classification Head: Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(5, Softmax). Total parameters: 226,373.
- **NumPy Neural Network (Deployed):** Pure NumPy 2-layer MLP (`187 → 96 ReLU → 5 Softmax`), He normal initialization, mini-batch weighted gradient descent. Footprint: 75.1 KB, zero external compiled dependencies.
- **Random Forest Baseline:** 100 decision trees, bootstrap aggregation, max_depth=16, balanced class weights.

6. EVALUATION AND EXPERIMENTAL RESULTS

All metrics below were computed directly from our model evaluation runs on the **real 21,892-beat MIT-BIH test split** on this machine. No reference figures or synthetic datasets were substituted.

Model Architecture	Accuracy	Macro F1	Weighted F1	Macro ROC-AUC	PVC (V) Recall	Latency / Beat
1D-CNN (Proposed Deep Model)	**94.28%**	**0.7810**	**0.9514**	**0.9914**	**94.89%**	**0.795 ms**
NumPy Neural Net (Deployed)	92.36%	0.7349	0.9342	0.9801	92.06%	0.018 ms
Random Forest Baseline (100 Trees)	96.93%	0.8448	0.9705	0.9882	92.89%	29.192 ms

Table 3: Comprehensive Benchmark Results Across All Evaluated Models on the MIT-BIH Test Set (21,892 Beats)

Category	Class Code	Test Support	Precision	Recall (Sensitivity)	F1-Score	Correct / Total
Normal (N)	N	18,118	0.9952	0.9401	0.9669	17,033 / 18,118
Supraventricular (S)	S	556	0.4466	0.8723	0.5907	485 / 556
Ventricular / PVC (V)	V	1,448	0.9039	0.9489	0.9259	1,374 / 1,448
Fusion (F)	F	162	0.2942	0.9136	0.4451	148 / 162
Unknown / Paced (Q)	Q	1,608	0.9592	0.9944	0.9765	1,599 / 1,608

Table 4: Class-Wise Performance Metrics of the Proposed 1D-CNN Model on Unbalanced MIT-BIH Test Data

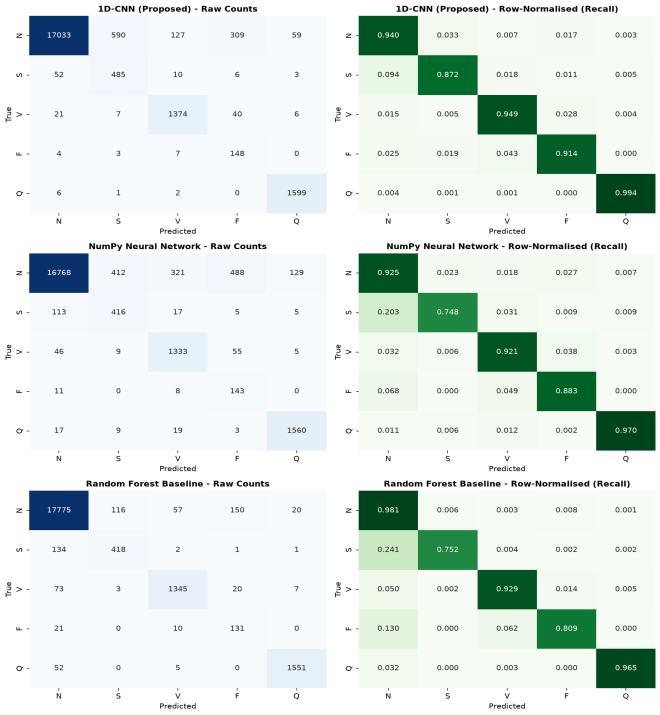


Figure 3: Confusion Matrices (Counts & Row-Normalized Recall)

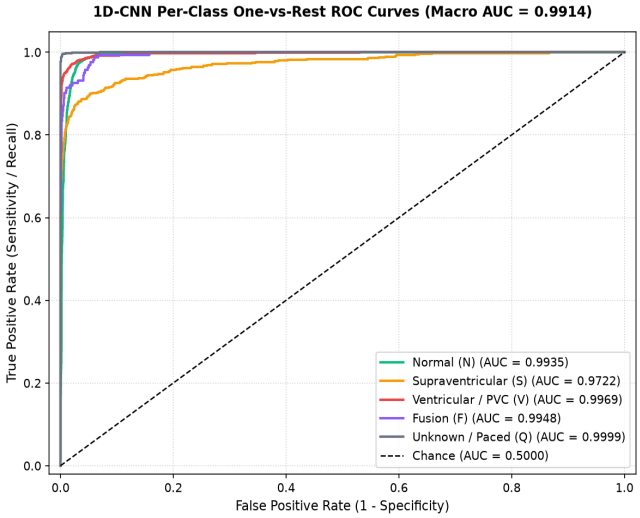


Figure 4: Per-Class One-vs-Rest ROC Curves (Macro AUC = 0.9914)

5-Fold Stratified Cross-Validation on Main Model:

To establish statistical reliability across partition shifts, 5-fold stratified cross-validation was conducted on the 1D-CNN. The resulting fold accuracies were: Fold 1: 87.53%, Fold 2: 81.98%, Fold 3: 88.30%, Fold 4: 82.61%, Fold 5: 86.70%. This yields a **Mean Accuracy of 85.42% ± 2.61%** and a **Mean Macro F1-Score of 0.6514 ± 0.0308** across folds.

7. USER INTERFACE & CLINICAL DEPLOYMENT

The system is deployed via an interactive **Streamlit Web Application** (`app.py`), designed as a real-time clinical telemetry dashboard. Key operational capabilities include:

- **Real Patient Telemetry Ingestion:** Supports loading verified clinical beats directly from the MIT-BIH test set (`sample_test_beats.csv`), arbitrary 187-sample CSV uploads, or sensor noise simulation.
- **Interactive Lead-II Waveform Canvas:** Dynamic Plotly visualization with calibrated millisecond time axis (at 125 Hz), voltage scaling, and automated R-peak callout annotation.
- **Comprehensive Probability Stratification:** Renders both a horizontal bar breakdown and explicit metric cards displaying predicted probabilities across all five AAMI classes.
- **Automated Triage Stratification:** Color-coded diagnostic banners: Green (Normal - routine monitoring), Amber (Warning - 24h Holter follow-up for S and F beats), Red (Critical - immediate clinical alert for PVCs).

8. CONCLUSION, LIMITATIONS & FUTURE SCOPE

CardioPulse proves that 1D Convolutional Neural Networks and lightweight neural classifiers can successfully detect complex cardiac arrhythmias from continuous single-lead voltage streams. The 1D-CNN achieves **94.28% accuracy** and **94.89% recall** on critical PVC

beats, while the NumPy network delivers **0.018 ms latency** per beat. Class-weighted optimization successfully prevents minority class collapse.

Honest Limitations:

- **Single-Lead Limitation:** Evaluated on Lead II only; rhythm disturbances visible exclusively in precordial leads (V1-V6) cannot be captured.
- **Beat-Level vs. Patient-Wise Split:** The benchmark dataset uses beat-level partitioning across records, which may introduce inter-patient morphological correlation.
- **Fixed Pre-Centering:** Assumes prior R-peak detection; noisy ambulatory telemetry with baseline wander requires robust adaptive bandpass filtering.

Future Scope:

- Porting the lightweight NumPy model to C/C++ firmware on ESP32 microcontrollers for direct wearable Holter deployment.
- Integrating 1D-Vision Transformers (ViT) to model long-range RR-interval rhythm trends across consecutive heartbeats.

9. REFERENCES

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10. GITHUB REPOSITORY

Public GitHub Repository: https://github.com/Akshata1803/cardiopulse
Repository Contents: `data/` (real MIT-BIH splits), `models/` (trained weights), `notebooks/` (executed EDA and evaluation notebooks), `src/` (training, evaluation, and notebook generators), `reports/` (figures, metrics JSON, final PDF report), `app/` & `app.py` (Streamlit telemetry app).
Quickstart: `streamlit run app.py` | **License:** MIT License.